Introduction:

Artificial Intelligence (AI) is the field of building systems that can perform tasks that normally require human-like intelligence.

Examples:

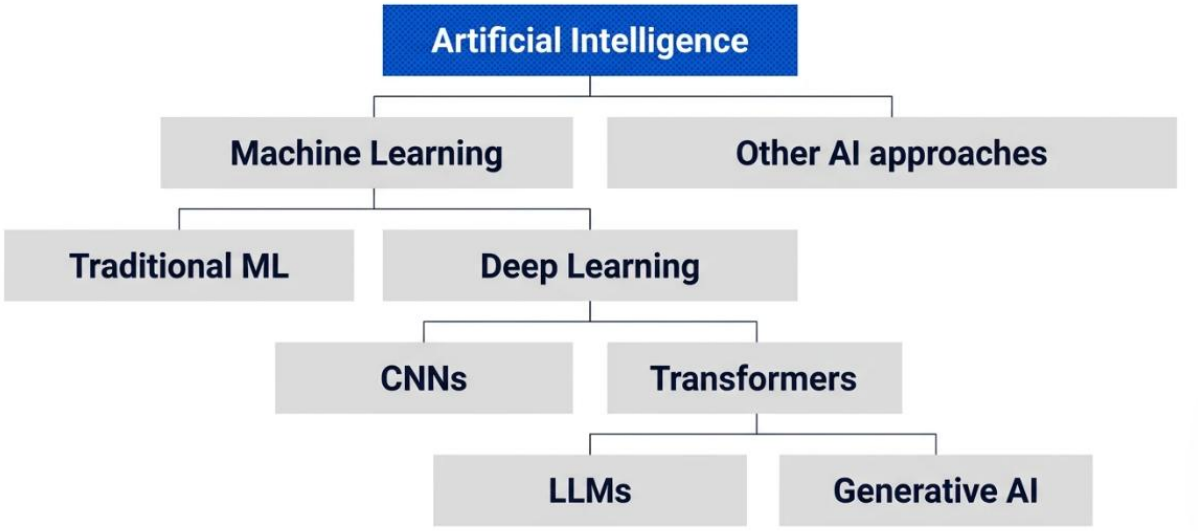
- Recognizing a face in a photo

- Understanding speech

- Recommending a movie

AI is the broad field.

Machine Learning, Deep Learning, NLP, Computer Vision, and Generative AI are areas within or closely associated with AI.



1)Machine Learning

A **subset of AI** where systems learn patterns from data rather than being explicitly programmed with every rule.

For example, instead of writing:

```
IF transaction > ₹50,000
AND location is unusual
AND many transactions happened quickly
THEN fraud
```

we can give a model many historical transactions labeled as:

```
Transaction → Fraud / Not Fraud
```

and let it learn patterns.

2)Deep Learning

A subset of machine learning based primarily on **multi-layer neural networks**.

It is particularly powerful for things such as:

Images, peech, natural language and complex pattern recognition

3)Traditional Programming vs Machine Learning

Traditional programming

You provide:

```
Rules + Data
    ↓
 Program
    ↓
 Output
```

For example, suppose you're writing a program to determine whether a student passes.

You might explicitly define:

```
IF marks >= 50
    PASS
ELSE
    FAIL
```

The programmer defines the rules.

Machine Learning

The approach changes:

Data + Expected Answers

↓

ML Model

↓

Learned Pattern

For example:

Study Hours	Attendance	Result
2	60%	Fail
5	80%	Pass
7	90%	Pass
1	50%	Fail

Table 1.1

The model tries to learn the relationship between the inputs and the result.

Then we can give it:

Study Hours = 6

Attendance = 85%

and ask it to predict: Pass

Key idea

Traditional programming: **Human writes the rules.**

Machine learning: **Model learns patterns from examples.**

What does **"learning"** mean?

Learning doesn't mean it thinks like human. It means that the model adjusts its **parameters** so that its predictions become better on the training data.

In table 1.1,the model looks for patterns such as longer study hrs → higher marks.

It learns numerical relationship represented by its parameters.

4) Dataset, Features and Labels

Hours Studied	Attendance	Assignments	Result
2	60	3	Fail
5	80	7	Pass
7	90	9	Pass
1	50	2	Fail

Dataset

The complete collection of examples.

The entire table is the **dataset**.

Features

The input information used by the model.

Here:

- Hours Studied
- Attendance
- Assignments

We can represent them as:

X = [Hours Studied, Attendance, Assignments]

Label / Target

The answer we're trying to predict. Here:

Result is the **label** or **target**.

X → Features

y → Target

For example:

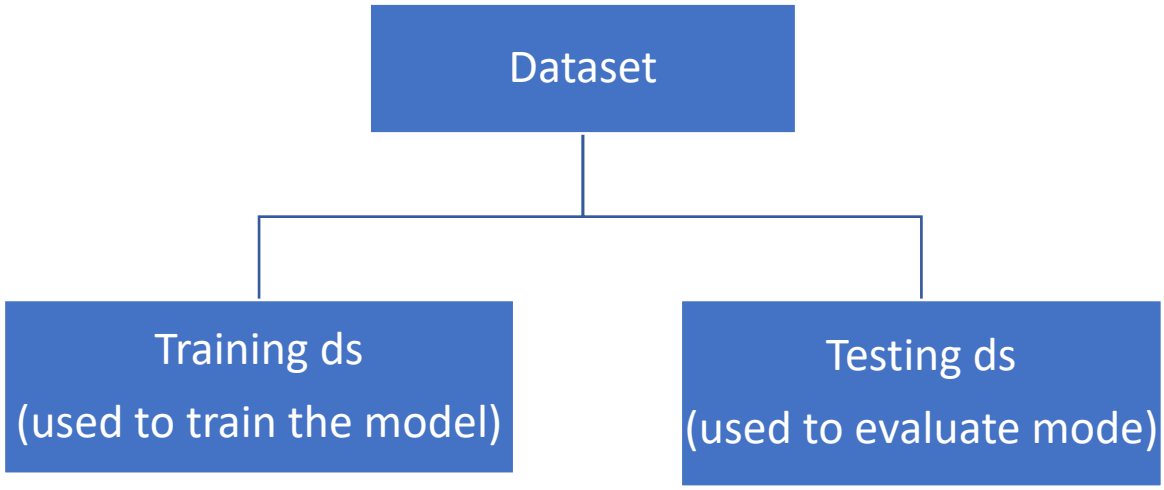
X = [5, 80, 7]

y = Pass

5. Training vs Prediction

Suppose we have 10,000 historical transactions.

We don't simply feed everything into a model and declare victory.



Training data: Used to learn patterns.

Test data: Used to see how well the trained model performs on unforeseen data.